

Qatar-1b

R-2798

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_130616 · created 2026-09-04T23:31:20+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456459.886933 +/- 5.9e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.138 +/- 0.049
Transit depth (Rp/Rs)^2	1.9 +/- 1.34 [%]
Orbital Inclination (inc)	81.72 +/- 2.54
Airmass coefficient 1 (a1)	0.65 +/- 0.011
Airmass coefficient 2 (a2)	0.35 +/- 0.16
Scatter in the residuals of the lightcurve fit is	1.73 %
Best Comparison Star	#2 - [539, 180]
Optimal Method	PSF photometry
Transit Duration (day)	0.043 +/- 0.026

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.138 ± 0.049 vs 0.1463 (deviation 0.17σ)
Duration fit vs published	0.043 d vs 0.0692 d (deviation 1.01σ)
Mid-time O-C	-0.1 min
Depth SNR	2.8
Residual scatter	1.73 %

Light curve

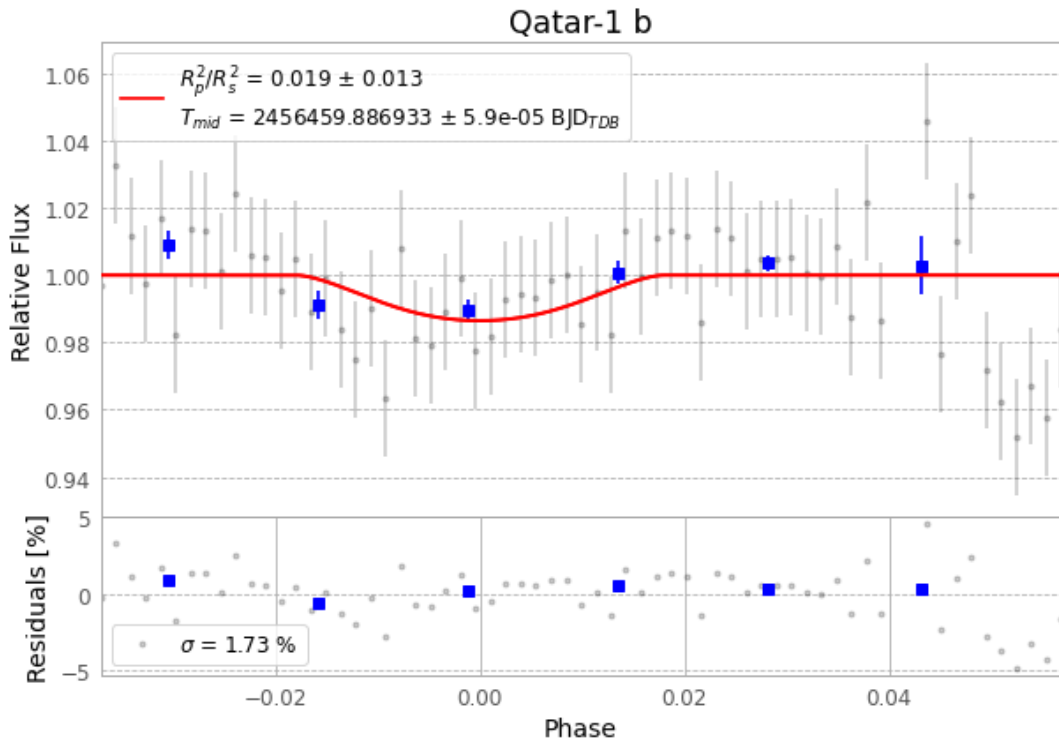


Figure 1 — FinalLightCurve_Qatar-1 b_2013-06-16.png (SHA-256 6484c7cae56c1a33...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [491, 236], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 36c7f50766cffe6b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

65 FITS frames (43 MB), Qatar-1130616080012.FITS → Qatar-1130616111212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-130616.log (b30b075bee25...), FinalLightCurve_Qatar-1 b_2013-06-16.png (6484c7cae56c...), FinalLightCurve_Qatar-1 b_2013-06-16.csv (0bebc118f3f8...)

Compute resources

Wall time 319.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-04 23:31Z.